

Building a Reproducible Scholarly Search Toolkit

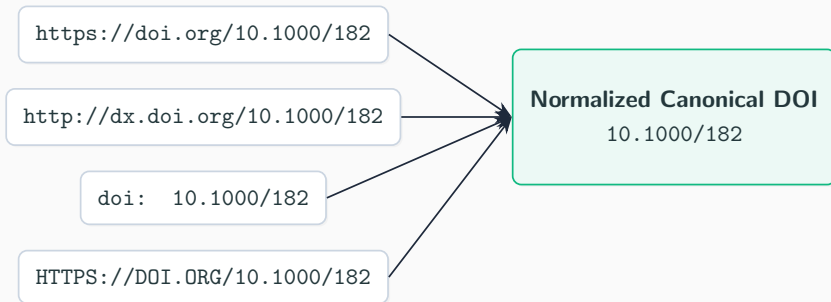
Episode 3: Persistent Identifiers (`ExternalIds`)

Nexus Scholar Suite

The Identifier Zoo

The Problem of Identity

A single paper may arrive with wildly differing identifier representations across databases.



The ExternalIds Contract

```
@dataclass
class ExternalIds:
    doi: str | None = None
    arxiv_id: str | None = None
    pubmed_id: str | None = None
    openalex_id: str | None = None
    s2_id: str | None = None

    def __post_init__(self) -> None:
        if self.doi:
            val = self.doi.strip().lower()
            for p in ("https://doi.org/", "http://doi.org/", "doi:"):
                if val.startswith(p): val = val[len(p):]
            self.doi = val.strip() or None
```

- **Case-Insensitive:** Lowercases all DOI strings deterministically.
- **Prefix Stripping:** URL schemes and URN prefixes are stripped.
- **Whitespace Cleansing:** Blank strings become None.
- **No Cross-Contamination:** Separate fields for DOI, arXiv, PubMed, OpenAlex, and S2 Corpus IDs.